

Experimental and Empirical Analysis of Heat Transfer in Cylinder Cross-Flow

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Abstract

This experiment aimed to measure the convective heat transfer characteristics of a heated cylinder in crossflow and compare the results to empirical models. Using forced convection, the heat transfer was analyzed by varying airflow velocity and evaluating dimensionless parameters such as the Reynolds and Nusselt numbers. The experimental results demonstrated an increase in the convective heat transfer coefficient and Nusselt number with higher airflow velocities and Reynolds numbers, consistent with theoretical expectations. However, deviations from empirical predictions were observed, highlighting the challenges posed by non-ideal flow conditions, uneven heat transfer, and insufficient time for thermal equilibrium. Despite these discrepancies, the experiment successfully met its objectives by measuring the heat transfer characteristics and evaluating them against empirical models, providing insights into the practical application of forced convection theory and the limitations of empirical correlations in experimental settings.

Introduction

Heat transfer is one of the fundamental processes that occurs throughout engineering. From the direct transfer of heat to radiation and convection, understanding how these mechanisms work is important to accurately and fully understand the behavior of systems and their interactions within themselves and the environment they are in. This experiment investigates the heat transfer characteristics from a heated cylinder in a variable velocity cross flow. The objective is to experimentally determine the mean convective heat transfer coefficient and compare it to predictions based on classic empirical models.

Forced convection is a method of heat transfer where a fluid is forced to flow over a surface. This method enhances the transfer of thermal energy between the fluid and the surface by disrupting the thermal boundary layer, the thin region near the surface where heat transfer occurs. Unlike natural convection, where fluid movement is driven by buoyancy forces due to temperature differences, forced convection relies on mechanical means to achieve higher heat transfer rates. The efficiency of forced convection depends on factors such as fluid velocity, fluid properties, and surface geometry. The convective heat transfer coefficient is highly dependent on flow conditions and is characterized by dimensionless numbers such as the Reynolds number and Nusselt number [4].

The Reynolds number is a dimensionless quantity used to predict the behavior of fluid flow in a given system. It represents the ratio of inertial forces to viscous forces in a fluid and is a critical parameter for determining whether the flow is laminar, transitional, or turbulent. In laminar flow, the fluid moves in smooth, orderly layers,

while in turbulent flow, the motion becomes chaotic and unpredictable. The Reynolds number depends on factors such as the fluid's velocity, its characteristic length (or the diameter of the cylinder in this experiment), and its viscosity [3].

The Nusselt number is a dimensionless parameter that quantifies the effectiveness of convective heat transfer relative to conductive heat transfer in a fluid. It is a measure of how efficiently heat is transported from a surface into the fluid by convection compared to conduction alone. A higher Nusselt number indicates more efficient convective heat transfer, often due to increased turbulence or flow velocity. The value of the Nusselt number depends on the flow conditions, surface geometry, and thermal properties of the fluid [2].

In this experiment, there are several assumptions made to simplify the calculations and derive useful results. It is assumed that the system reaches steady-state conditions, where the surface temperature of the cylinder, ambient air temperature, and heat input remain constant over time, ensuring that the heat generated by the cylinder is entirely transferred to the airflow. Additionally, the airflow in the duct is considered uniform, with a consistent velocity profile and temperature distribution across the cylinder. Heat transfer is presumed to be dominated by convection, with radiative and conductive losses considered negligible. To account for variations in fluid properties with temperature, the thermal and physical properties of air are interpolated at the mean film temperature, treating these variations as linear within the experimental range. These assumptions help streamline the calculations while maintaining reasonable accuracy in the experimental results.

This experiment uses a heated cylindrical element placed perpendicular to the airflow in the test section of a vertical wind tunnel. The temperature of the rod is maintained at 95°C by increasing the supplied voltage as the airflow across the rod increases. Measurements of surface and ambient temperature and flow pressure are recorded. The velocity of the airflow is varied to observe the relationship between the convective heat transfer coefficient and the flow characteristics. From the measured temperature and pressure, the flow velocity can be calculated using equation (). Reynolds and Nusselt's numbers are also calculated to compare experimental results to empirical values.

Equations

Flow Velocity with Derivation Starting From Bernoulli's Equation

$$P_1 + \frac{1}{2}\rho U_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho U_2^2 + \rho g h_2$$

Assume h is constant, assume $U_2 = 0$

$$P_1 + \frac{1}{2}\rho U_1^2 = P_2$$

Solve for velocity,

$$U_1 = \sqrt{\frac{2(P_1 - P_2)}{\rho}}$$

Pressure difference using a manometer,

$$P_1 - P_2 = \rho g H$$

Substitute into velocity,

$$U_1 = \sqrt{\frac{2(\rho_m g H)}{\rho}}$$

Using ideal gas law,

$$\rho = \frac{P_a}{RT_a}$$

Substitute into velocity,

$$U_1 = \sqrt{\frac{2(\rho_m g H) RT_a}{P_a}}$$

Simplifying numerator using $\rho_m = 1000 \text{ kg/m}^3$, $g = 9.81 \text{ m/s}^2$, and $R = 287 \text{ J/kgK}$,

$$U = 75 \sqrt{\frac{T_a H}{P_a}} \quad (1)$$

Average Nusselt Number, Empirical Relation

$$\overline{Nu}_D = C * Re_D^m * Pr^{1/3} \quad (2)$$

Nusselt Number

$$Nu = \frac{hD}{k_f} \quad (3)$$

Film Temperature

$$T_{film} = \frac{T_a + T_s}{2} \quad (4)$$

Thermal Conductivity

$$k_f = 26.3 * 10^{-3} \left(\frac{300 - T_{film}}{300 - 350} \right) + (26.3 * 10^{-3} - 30 * 10^{-3}) \quad (5)$$

Kinematic Viscosity

$$\nu_{film} = 15.89 * 10^{-6} \left(\frac{300 - T_{film}}{300 - 350} \right) + (15.89 * 10^{-6} - 20.92 * 10^{-6}) \quad (6)$$

Prandtl Number

$$Pr = 0.707 - \left(\frac{300-T_{film}}{300-350}\right)(0.707 - 0.700) \quad (7)$$

Kinematic Viscosity

$$\nu_a = 11.4 * 10^{-6} \left(\frac{250-T_{film}}{250-300}\right) + (11.44 * 10^{-6} - 15.89 * 10^{-6}) \quad (8)$$

Kinematic Viscosity

$$\nu_s = 20.92 * 10^{-6} \left(\frac{350-T_{film}}{350-400}\right) + (20.92 * 10^{-6} - 26.41 * 10^{-6}) \quad (9)$$

Heat Transfer Rate

$$q = \bar{h}A_s(T_s - T_a) \quad (10)$$

Reynolds Number

$$Re_D = \frac{UD}{\nu_a} \quad (11)$$

Power Relationship

$$q = P = \frac{V^2}{R} \quad (12)$$

Variables

U = Flow Velocity (m/s)

H = Manometer Height (mm H_2O)

P_a = Ambient Pressure (Pa) = 101325

$\overline{Nu}_D$ = Average Nusselt Number, Ratio of Convective to Conductive Heat Transfer

C = Empirical Constant, Dependent on Geometry and Flow Regime = 0.193

Re_D = Reynolds Number Using Cylinder Diameter, Ratio of Inertial Forces to Viscous Forces

m = Empirical Exponent, Based on Experimental Data = 0.618

Pr = Prandtl Number, Ratio of Momentum Diffusivity to Thermal Diffusivity

Nu = Nusselt Number

h = Convective Heat Transfer Coefficient (W/m^2K)

$\bar{h}$ = Mean Convective Heat Transfer Coefficient (W/m^2K)

k_f = Thermal Conductivity of the fluid (W/mK)

T_{film} = Film Temperature, Average Temperature Between the Surface and the Ambient Air (K)

T_a = Ambient Air Temperature (K)

T_s = Surface Temperature of the Cylinder (K)

ν_{film} = Kinematic Viscosity of the Fluid (m^2/s), General

ν_a = Kinematic Viscosity of the Fluid (m^2/s), Temperature Range of 250 to 300 Kelvin

ν_s = Kinematic Viscosity of the Fluid (m^2/s), Temperature Range of 350 to 400 Kelvin

q = Heat Transfer Rate (W)

A_s = Surface Area of the Cylinder (cm^2) = 24.82

D = Cylinder Diameter (cm) = 5

V = Voltage Reading (Volts)

R = Heater Resistance (Ohms) = 70.9

Experimental Setup

This experiment consists of a heated cylindrical rod placed perpendicularly in a controlled airflow duct to analyze forced convection heat transfer. The duct has a fan at the top to regulate the airflow velocity, which can be adjusted using an iris damper as seen in Appendix A.2 and A.3. The velocity of the air is indirectly measured through the pressure differential recorded by a manometer connected to the duct, located above the test section. The rod is heated by passing an electric current through a cartridge heater, and its surface temperature is monitored using a thermocouple. The ambient air temperature is also measured to establish the temperature gradient necessary for calculating heat transfer.

The cylindrical rod has a circumference of 5 cm and an exposed surface area of 24.82 cm². The thermocouples are connected to an instrument console, which displays the surface temperature (T_s) and the ambient temperature (T_a). To ensure steady-state conditions, the rod's surface temperature is maintained at approximately 95°C by adjusting the heat input through the console and being allowed to settle in equilibrium. The fan speed is increased incrementally to cover a range of duct velocities, and at each speed, the steady-state temperatures, voltage, and pressure differential are recorded for analysis using the MATLAB code found in Appendix B.

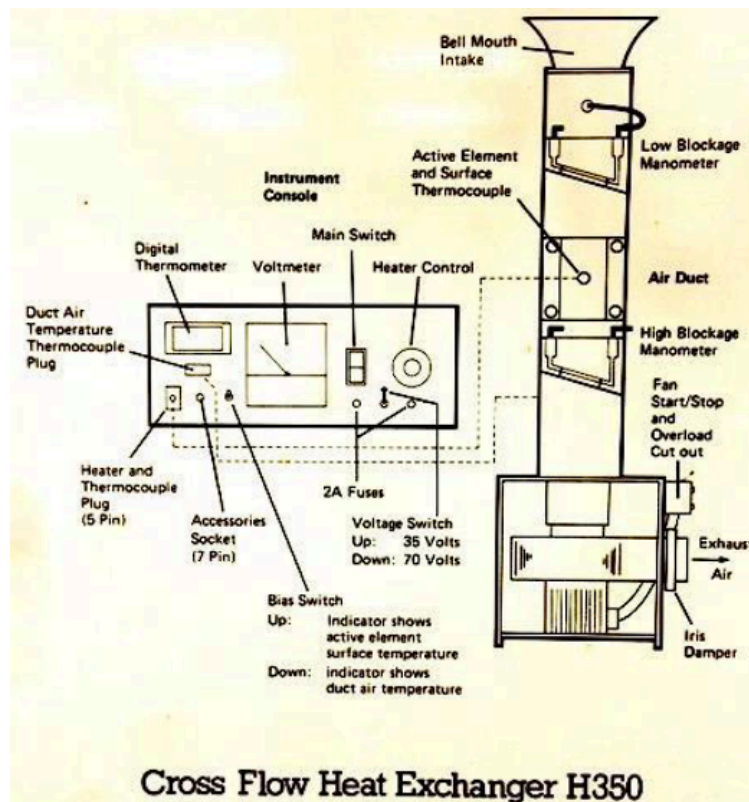


Figure (1) Schematic of PA Hilton H350 Cross Flow Heat Exchanger. For this experiment, the high blockage manometer is not in use

The schematic diagram of the experimental setup is provided to illustrate the arrangement of the duct, cylinder, manometer, and thermocouples. The airflow is assumed to be uniform, and the experimental conditions are maintained as steady-state. Steady-state eliminates the need to account for any transient heat transfer. As a result, the system can be considered in equilibrium, where all temperatures and rates remain constant. Data collected during the experiment is used to calculate the convective heat transfer coefficient and compare it to predictions from empirical correlations. The instrument console allows for accurate monitoring and adjustments, while the manometer readings provide a basis for calculating the duct velocity using equation (1).

Results

This data analysis aims to calculate and compare the experimental and empirical heat transfer coefficients around a heated cylinder in cross-flow. The calculations are based on steady-state temperature readings, airflow velocity measurements, and pressure readings. The following tables summarize the key results obtained and calculated from this experiment. See Appendix B for the MATLAB code containing all the calculations done, as well as the raw data taken from the experiment.

Duct Velocity

Using equation (1), which is derived from Bernoulli's equation, along with the ambient temperature and pressure readings, and manometer readings, the velocity of the flow is calculated for each iris damper setting. This is later used in Figures (1) and (3) to show the increase in heat transfer coefficient with the increase in airflow velocity. The values are tabulated in Table (1).

	Orifice 8	Orifice 7	Orifice 6	Orifice 5	Orifice 4	Orifice 3	Orifice 2	Orifice 1	Orifice 0	Orifice 6.5
Velocity (m/s)	11.812	16.206	19.641	22.558	24.98	26.88	28.797	30.32	30.455	19.004

Table (1) Tabulated air velocity values based on equation (1).

Experimental Heat Transfer Coefficient

Using equation (10), the experimental heat transfer coefficients were calculated. The active element area, the temperature difference between the surface and ambient, and equation (12) were used. This result is seen in Figures (1) and (3) along with the empirical coefficients as a function of flow velocity. The values are tabulated in table (2)

	Orifice 8	Orifice 7	Orifice 6	Orifice 5	Orifice 4	Orifice 3	Orifice 2	Orifice 1	Orifice 0	Orifice 6.5
Heat Transfer Coefficient	125.24	141.77	158.07	173.38	184.63	193.12	195.95	208.17	211.65	151.75

Table (2) Tabulated experimental heat transfer coefficient values based on equation (10).

Reynolds Number

Using equation (11), along with the flow velocities, element diameter D , and equation (9), the Reynolds number was calculated at each iris damper setting. It is then used in Figure (2) to show the relationship between the Nusselt number and Reynolds number, both experimental and empirical.

	Orifice 8	Orifice 7	Orifice 6	Orifice 5	Orifice 4	Orifice 3	Orifice 2	Orifice 1	Orifice 0	Orifice 6.5
Reynolds Number	9924.2	13624	16492	18964	20987	22603	24187	25488	25586	15971

Table (3) Tabulated Reynolds number values based on equation (11). The characteristic length used in the diameter of the cylinder.

Empirical Heat Transfer Coefficient

Using equations (2), (3), (4), (5), (7), and (11), the empirical heat transfer coefficients were calculated. They are then compared to the experimental coefficients as seen in Figures (1) and (3) as functions of flow velocity. The calculated values are tabulated in Table (4)

	Orifice 8	Orifice 7	Orifice 6	Orifice 5	Orifice 4	Orifice 3	Orifice 2	Orifice 1	Orifice 0	Orifice 6.5
Heat Transfer Coefficient	91.197	110.89	124.85	136.04	144.87	151.61	158.17	163.31	163.75	122.35

Table (4) Tabulated empirical heat transfer coefficient values based on equations (2), (3), and (5).

Experimental Nusselt Number

Using equations (3) and (5), along with the experimental heat transfer coefficients and element diameter D , the Nusselt number was calculated for each iris damper setting. This is compared to the empirical Nusselt number in Figure (2). The calculated values are tabulated in Table (5)

	Orifice 8	Orifice 7	Orifice 6	Orifice 5	Orifice 4	Orifice 3	Orifice 2	Orifice 1	Orifice 0	Orifice 6.5
Nusselt Number	69.528	78.727	87.732	96.282	102.5	107.26	108.77	115.6	117.5	84.255

Table (5) Tabulated experimental Nusselt number values based on equation (3).

Empirical Nusselt Number

Using equations (2), (4), (7), (9), and (11), the empirical Nusselt number was calculated for each iris damper setting. The coefficients for equation (2) were found based on the calculated Reynolds numbers and found from published results [2]. This is

compared to the experimental Nusselt number in Figure (2). The calculated values are tabulated in Table (6).

	Orifice 8	Orifice 7	Orifice 6	Orifice 5	Orifice 4	Orifice 3	Orifice 2	Orifice 1	Orifice 0	Orifice 6.5
Nusselt Number	50.629	61.58	69.297	75.544	80.428	84.202	87.8	90.69	90.906	67.936

Table (6) Tabulated empirical Nusselt number values based on equation (2).

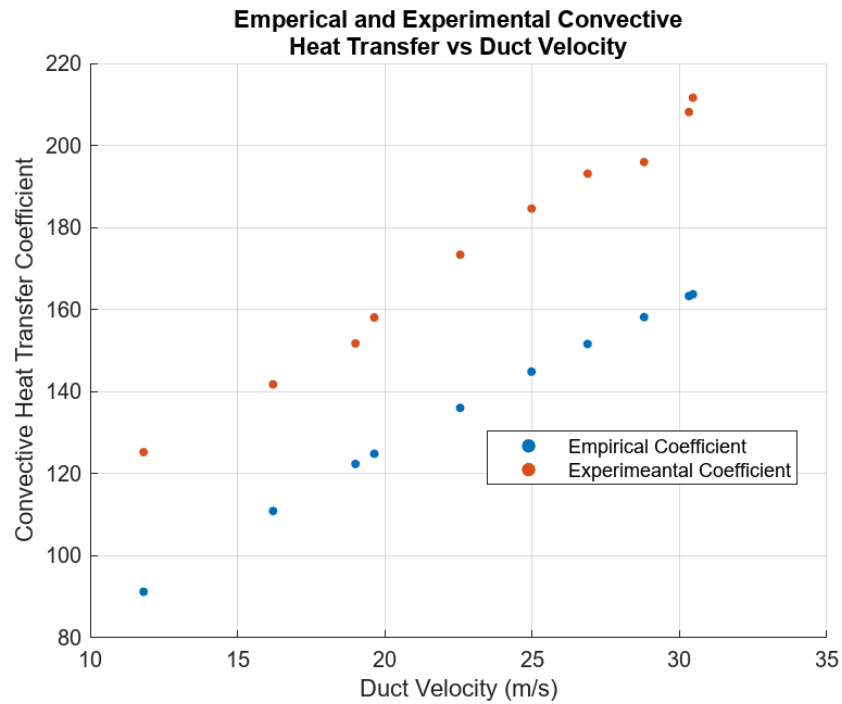


Figure (1) Comparison between experimental and empirical values of convective heat transfer coefficients vs the airflow velocity.

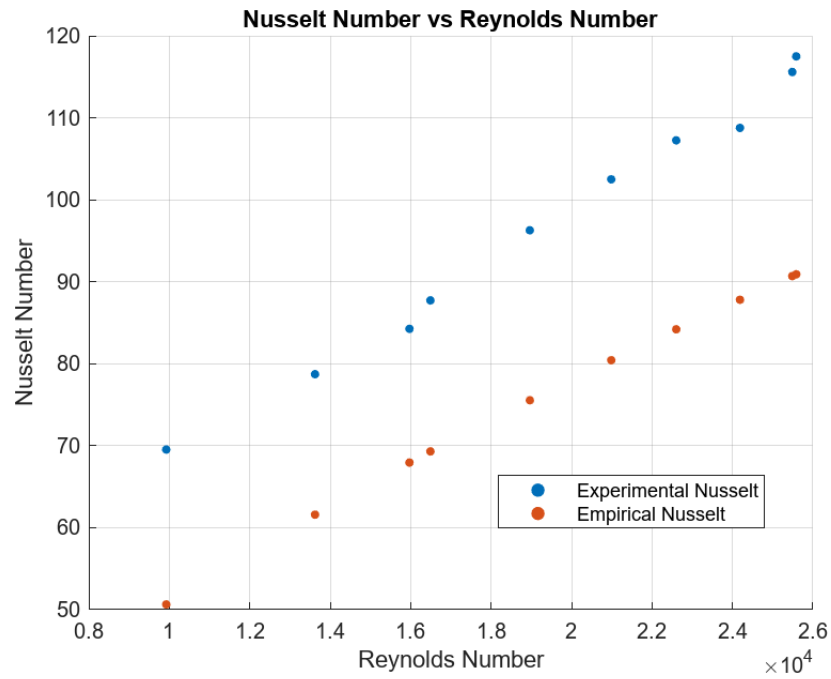


Figure (2) Comparison between experimental and empirical Nusselt numbers vs the Reynolds numbers.

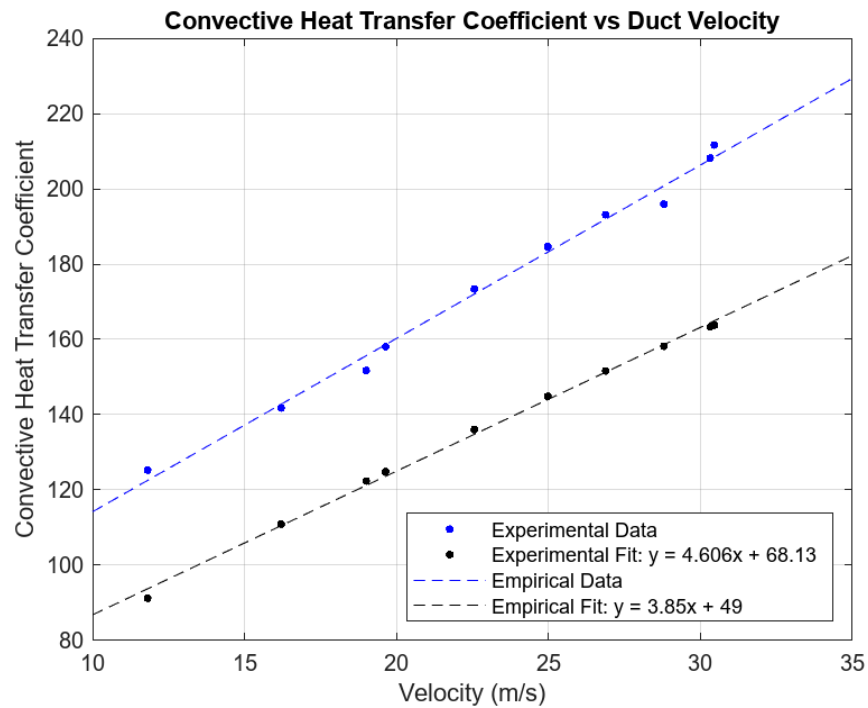


Figure (3) Comparison between experimental and empirical values of convective heat transfer coefficients vs the airflow velocity with linear lines of best fit.

Discussion of Results

The empirical correlation used for the Nusselt number,

$$\overline{Nu}_D = C * Re_D^m * Pr^{1/3}$$

is based on dimensional analysis and extensive experimental studies. It is named after Wilhelm Nusselt, a German Engineer who pioneered much of our early understanding of convective and conductive heat transfer. Later, researchers such as Zukauskas refined this correlation by conducting experiments on cylinders in crossflow across a wide range of Reynolds and Prandtl numbers. Their experiments involved the precise measurement of heat transfer rates and allowed them to fit constants C and m depending on the Reynolds number used. The above equation is widely used in engineering and is one of the foundational equations in heat transfer for designs involving cylindrical geometry [1].

The experimental and empirical convective heat transfer coefficients were calculated and compared across a range of flow velocities. The empirical heat transfer coefficient, calculated using the widely accepted Nusselt number correlation found in equation (2), matched expectations for flow over a circular cylinder in crossflow. However, the experimental results showed significant deviation as seen in Figures (1) and (3), which can be attributed to several factors. Potential sources of error include inaccuracies in temperature measurements from not waiting long enough for equilibrium to be achieved, heat loss due to radiation or conduction, and non-ideal flow conditions in the duct. If the temperature was not allowed to reach equilibrium, that error would propagate throughout all the calculations to get the heat transfer coefficients, as all the equations are functions of the difference between the surface and ambient temperatures. If the flow is not ideal, then error could result from the instability caused by uneven flow. There could be uneven heat transfer around the cylinder, meaning that different parts of the cylinder experience varying heat removal rates, leading to inaccuracies in the calculated heat transfer coefficients and Nusselt numbers. Additionally, the empirical correlation assumes idealized conditions that may not align with this experimental setup. While there is a deviation from empirical results, the deviation is consistent across all the calculated data, suggesting that the error is from the experimental setup.

Conclusion

The experiment successfully compared the experimental and empirical convective heat transfer coefficients and Nusselt numbers for a heated cylinder in crossflow. Both experimental and empirical coefficients followed the expected trends

of increasing with flow velocity and Reynolds number. However, significant deviations were observed between the two, as highlighted in Figures (1) and (3). These discrepancies are primarily attributed to errors in temperature measurements caused by insufficient time for equilibrium to be achieved, heat losses due to radiation or conduction, and non-ideal flow conditions within the duct. Additionally, uneven heat transfer around the cylinder may have led to inaccuracies in the calculated heat transfer coefficients and Nusselt numbers. These results demonstrate the challenges of aligning real-world experimental setups with the idealized conditions assumed in empirical correlations. The experiment effectively highlighted the applicability of forced convection theory and the relationship between flow velocity and heat transfer. However, issues such as temperature equilibrium and non-ideal flow conditions emphasize the need for improved experimental control and more precise measurement techniques to reduce discrepancies and limit error propagation through calculation.

References

- [1] A. Zhukauskas and J. Žiugžda, *Heat Transfer of a Cylinder in Crossflow*. Core/Mechanical, 1985.
- [2] Y. A. Çengel, *Heat Transfer: a practical approach*. Boston (Mass.); Burr Ridge (Ill.); Dubuque (Iowa): Mcgraw-Hill, 2006.
- [3] J. Anderson, *Fundamentals of aerodynamics*, 6th ed. Mcgraw-Hill, 2016.
- [4] "Forced convection - Energy Education," *Energyeducation.ca*, 2018.
https://energyeducation.ca/encyclopedia/Forced_convection

4Appendix A

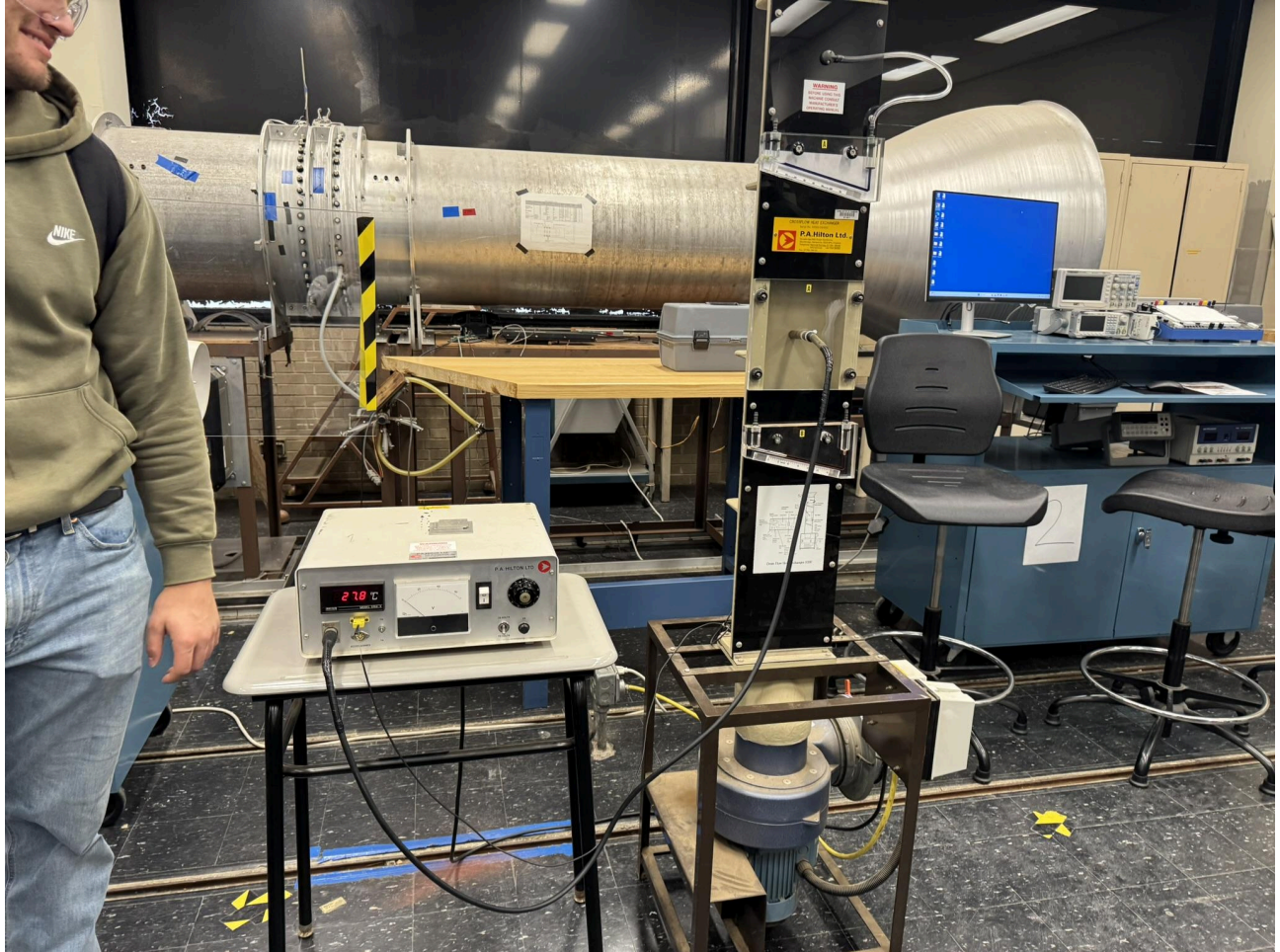


Figure (1) PA Hilton 350 Cross Flow Heat Exchanger. Voltage control and temperature readings are on the left, and the wind tunnel with cylindrical heat element is on the right. The manometer used is at the top of the wind tunnel.



Figure (2) Wind tunnel exit iris orifice with variable aperture. Acts as a damper to control the air velocity through the test section.



Figure (3) Iris damper settings. 0 is the most open and results in the highest flow rate. 8 is the most closed and results in the lowest flow rate.



Figure (4) Back of the wind tunnel. The thermocouple used to measure freestream flow is located on the bottom.

Appendix B

```

oriface = [8 7 6 5 4 3 2 1 0 6.5]; % 0 is open the most
H = [8.5 16 23.5 31 38 44 50.5 56 56.5 22]; % mm of water
T_s_c = [95.3 95.1 95.5 95.1 95.2 94.9 95.3 95.1 95.3 95.2]; % Surface temp in
Celsius
T_s_k = T_s_c + 273;
T_a_c = [22.7 22.7 22.7 22.7 22.8 22.8 22.8 22.7 22.7 22.7]; % Ambient temp in
Celsius
T_a_k = T_a_c + 273;
volts = [40 42.5 45 47 48.5 49.5 50 51.5 52 44]; % volts

ambient_pressure = 101325; % Pa
R = 70.9; % Ohms
active_area = 24.82; % cm^2
active_area = active_area / 100^2; % m^2
circumference = 5; % cm
circumference = circumference / 100; % m
diameter = circumference / pi; % m

velocity = 75 * sqrt((T_a_k .* H) / ambient_pressure); % m/s

names = {'Orifice 8', 'Orifice 7', 'Orifice 6', 'Orifice 5', 'Orifice 4', 'Orifice
3', 'Orifice 2', 'Orifice 1', 'Orifice 0', 'Orifice 6.5'};
velocity_table = array2table(velocity, "VariableNames", names, "RowNames",
{'Velocity (m/s)'});
disp(velocity_table)

q = volts.^2 / R; % = P

T_film = (T_s_k + T_a_k) / 2;

air_viscosity = 11.4*10^-6 - ((250 - T_film) / (250 - 300)) * (11.44*10^-6 -
15.89*10^-6);
air_viscosity2 = 20.92*10^-6 - ((350 - T_film) / (350 - 400)) * (20.92*10^-6 -
26.41*10^-6);

Reynolds_diameter = (velocity .* diameter) ./ air_viscosity2;
reynolds_table = array2table(Reynolds_diameter, "VariableNames", names, "RowNames",
{'Reynolds Number'});
disp(reynolds_table)

% From Hilpert Correlation
C = 0.193;
m = 0.618;

h_experimental = q ./ (active_area * (T_s_k - T_a_k));
h_exp_table = array2table(h_experimental, "VariableNames", names, "RowNames",
{'Heat Transfer Coefficient'});

```



```

disp(h_exp_table)

Prandtl = 0.707 - ((300 - T_film) / (300 - 350)) * (0.707 - 0.7);

Nusselt_emperical = C .* Reynolds_diameter.^m .* Prandtl.^(1/3);
Nusselt_emp_table = array2table(Nusselt_emperical, "VariableNames", names,
"RowNames", {'Nusselt Number'});
disp(Nusselt_emp_table)

k_f = 26.3*10^-3 - ((300 - T_film) / (300 - 350)) * (26.3*10^-3 - 30*10^-3);

h_empirical = (Nusselt_emperical .* k_f) / diameter;
h_emp_table = array2table(h_empirical, "VariableNames", names, "RowNames", {'Heat
Transfer Coefficient'});
disp(h_emp_table)

Nusselt_experimental = (h_experimental .* diameter) ./ k_f;
Nusselt_exp_table = array2table(Nusselt_experimental, "VariableNames", names,
"RowNames", {'Nusselt Number'});
disp(Nusselt_exp_table)

figure(1)
scatter(velocity, h_empirical, 15, 'Filled')
hold on
grid on
scatter(velocity, h_experimental, 15, 'Filled')
hold off
title({'Emperical and Experimental Convective', 'Heat Transfer vs Duct Velocity'})
xlabel('Duct Velocity (m/s)')
ylabel('Convective Heat Transfer Coefficient')
legend('Empirical Coefficient', 'Experimeantal Coefficient', 'Location', 'Best')

figure(2)
scatter(Reynolds_diameter, Nusselt_experimental, 15, 'Filled')
hold on
grid on
scatter(Reynolds_diameter, Nusselt_emperical, 15, 'Filled')
hold off
title('Nusselt Number vs Reynolds Number')
xlabel('Reynolds Number')
ylabel('Nusselt Number')
legend('Experimental Nusselt', 'Empirical Nusselt', 'Location', 'Best')

[xData1, yData1] = prepareCurveData(velocity, h_experimental);
[xData2, yData2] = prepareCurveData(velocity, h_empirical);

ft = fittype('poly1');

```

```

[fitresult1, gof1] = fit(xData1, yData1, ft);
[fitresult2, gof2] = fit(xData2, yData2, ft);

figure(3);
plot(xData1, yData1, '.b', 'MarkerSize', 10);
hold on
plot(xData2, yData2, '.k', 'MarkerSize', 10);
plot(fitresult1, 'b--');
plot(fitresult2, 'k--');
title('Convective Heat Transfer Coefficient vs Duct Velocity');
xlabel('Velocity (m/s)');
ylabel('Convective Heat Transfer Coefficient');
grid on
legend('Experimental Data', 'Experimental Fit: y = 4.606x + 68.13', ...
      'Empirical Data', 'Empirical Fit: y = 3.85x + 49', 'Location', 'Southeast');
hold off

```

Figure (1) Complete MATLAB script used for all calculations. Raw data is listed at the top of the script. Output numbers and figures can be found in the results section.